



Project Rescue: A Literacy and Numeracy Enhancement



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Name of Project : ***Project Rescue: A Literacy and Numeracy Enhancement***

Proponent : MARY ANN L. DECREPITO

Implementing Agency : CapSU Pontevedra Campus –College of Education Arts and Sciences; Extension Department/Office of the Students' Affair

Collaborating Agency : Pontevedra Elementary School (DepEd)

Title of Campus Extension: KALAPIT

Program It Belongs : LITERACY

Recipient : Non-readers and Non-Numerates

Target No. of Recipients : One Hundred Forty Eight (148)

Duration : Three (3 months)

Target Date : March 2023

Total Cost : P 160,000.00

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I. Rationale

The existence of non-readers and non-numerates in every public elementary school had become a perennial problem of the Department of Education. Thus, a number of schools have had non-readers and non-numerates from Grade 1 to Grade 6. This is even become severe because of our precarious situation during the Covid 19 pandemic where education had been challenged. Schools find ways just to deliver education through modular learning. Indeed, an abrupt change of learning modes had affected the numeracy and reading fluency of the children. Stanford Graduate School of Education (GSE, 2021) as cited by Dagmil (2022) on her Project LETRA stated that students' development of oral reading fluency largely stopped in spring 2020 after the abrupt school closures brought on by COVID-19. Gains in these skills were stronger in fall 2020, but not enough to recoup the loss students experienced.

Moreover, poor numeracy is affecting the life chances of learners. It is not a new phenomenon in the elementary mathematics class. Many pupils still cannot perform the basic arithmetic which is essential part of their everyday life. For pupils to develop confidence and be successful learners they need to think creative and to communicate well. Thus, letting every learner expose to a challenging experience both in English and Mathematics would propel and motivate him to learn.

In line with the implementation of Basic Education Curriculum, the Department of Education (DepEd) implemented "Every Child A Reader Program" (ECARP), through DepEd Memorandum No.402.s.2004 and Administrative Order No. 324, hence public schools aimed to teach public elementary pupils with planned training in reading to make them independent young readers. This is supported with DO 18, S. 2017: Guidelines on the Utilization of the 2017 Every Child a Reader Fund for the Early Language, Literacy, and Numeracy Program: Professional Development Component, a Literacy and Numeracy Roadmap was implemented.

Furthermore, Pontevedra Elementary School in Pontevedra District is the priority district in Capiz Division for they ranked to be the bottom 3 during the pre-assessment in reading and numeracy conducted by the Capiz Division with 148 pupils who are non-readers and non-numerates. This alarming situation made the institution request for an extension program in remediation to Capiz State University- Pontevedra Campus.

Capiz State University as a learning institution with quadruple functions is dedicated to serve the community through its extension program particularly in the College of Education, Arts and Sciences. With L. I. T. E. R. A. C.Y. as its banner program it is deemed necessary to deliver educational services. Thus, with the advocacy to help the non-readers and non-numerates to upgrade their numeracy and reading level if not to be eradicated, "Project Rescue: A Literacy and Numeracy Enhancement" was launched. This enables the pupils to enhance reading and numeracy skills, make reading a habit and to have a well-conditioned childhood. It includes tutorial sessions, storytelling, interactive activities, Zumba dance, gift-giving and feeding program.

II. Objectives:

Generally, this extension project aims to enhance the reading and numeracy skills of the non-numerates and non-readers; develop their love for mathematics and reading and design a remediation tool both in numeracy and reading.

Specifically, it aims to:

1. conduct tutorial sessions both in reading and numeracy to the adopted school;
2. enhance the reading and numeracy level of the beneficiaries;
3. appreciate reading through differentiated instruction;
4. appreciate the significance of numbers in real-life;
5. make reading a habit;
6. craft a remedial reading workbook;
7. design a teacher's guide on beginning reading using the Oral Graphic Symbolic Language (OGSL)
8. craft a workbook with digital interactive exercises and manipulatives in basic Mathematics;
9. construct a reading and mathematics center (Hulot sang Kaalam)

III. Project Description

Pontevedra Elementary School is a central school located on the heart of the municipality. However, mostly of the pupils comprises all of the neighboring barangays and even coastal barangays which only has a primary school. This school is the priority district in the Division of Capiz when it comes to reading and numeracy for, they belong to the bottom three among the schools in the Division of Capiz as stated by the education program supervisors of both the English and Mathematics Department

Project Rescue: A Literacy and Numeracy Enhancement is an extension project under the LITERACY Program of the College of Education, Teacher Department. This training project was anchored on the research study "Books: Open, Flip and Read for a Positive Change" and from the request made by Pontevedra Elementary School.

This study was conducted to 148 beneficiaries from Grade 1 to Grade 6 pupils. A secondary data based from the result of the DepEd Capiz Division Beginning of School Year Comprehensive Rapid Literacy Assessment (BoSY CRLA) and from the result of the Philippine Informal Reading Inventory (Phil IRI) of Pontevedra Elementary School were utilized in selecting the non-readers and non-numerates. Likewise, a pre-test in reading and numeracy was given to validate the secondary data. Differentiated instruction such as games, storytelling, peer and individual tutoring were employed during the launching and tutorial sessions. Adopt-a-child scheme was employed during the 3-month everyday tutorial sessions employing the Oral Graphic Symbolic Language (OGSL).

Venue	Beneficiaries per Grade Level	DATE	NO. OF PARTICIPANTS		Total
			Non-Readers	Non-Numerates	
Pontevedra Elementary School	Grade 1	March 2023 to June 2023	24	24	24
	Grade 2		23	23	23
	Grade 3		41	41	41
	Grade 4		22	22	22
	Grade 5		23	23	23
	Grade 6		15	15	15
TOTAL			148		

This study was anchored on the Literacy theory of how people develop through various means such as the ability to use several core skills, including decoding, comprehension, writing, and speaking skills. This subject was important for educators because children need literacy skills to learn properly, and adults with low literacy rates often struggle economically. The understanding of this theory enables educators to consider the ways children were exposed to and interact with language. Educators apply this knowledge to enhance teaching ability and support literacy learning. Thus, both the non-readers and non-numerates were minimized or even eradicated.

Furthermore, some theories were also considered in this study. Such as the Constructivism Theory that focused on the skills reflected by children as they engage with the world and move through developmental stages; Ecological/Contextual Theories focused on the broader context of the child, while Sociocultural/Cooperative Theories also focused on the immediate surroundings of the child how they interact. Thus, using these theories gave a framework and builds our scientific understanding of how children grow and develop.

This three-month everyday tutorial sessions had its kick-off last March 2023 with the MOA signing both of the two parties the Capiz State University and DepEd Capiz Division. Post assessment both in reading and numeracy was conducted last June 2023 but the project continued so to enhance the reading ability of the respondents with the use of the designed outputs like the enhanced beginning reading and writing remedial workbook, reading worksheets, numeracy worksheets and tools with digital interactive exercises and manipulatives in basic Mathematics. A teacher guide for beginning reading using the Oral Graphic Symbolic Language (OGSL) was designed to be utilized and adopted by the BEEd students for further used on the conduct of reading tutorials and may consider by the DepEd teachers on their remedial class.

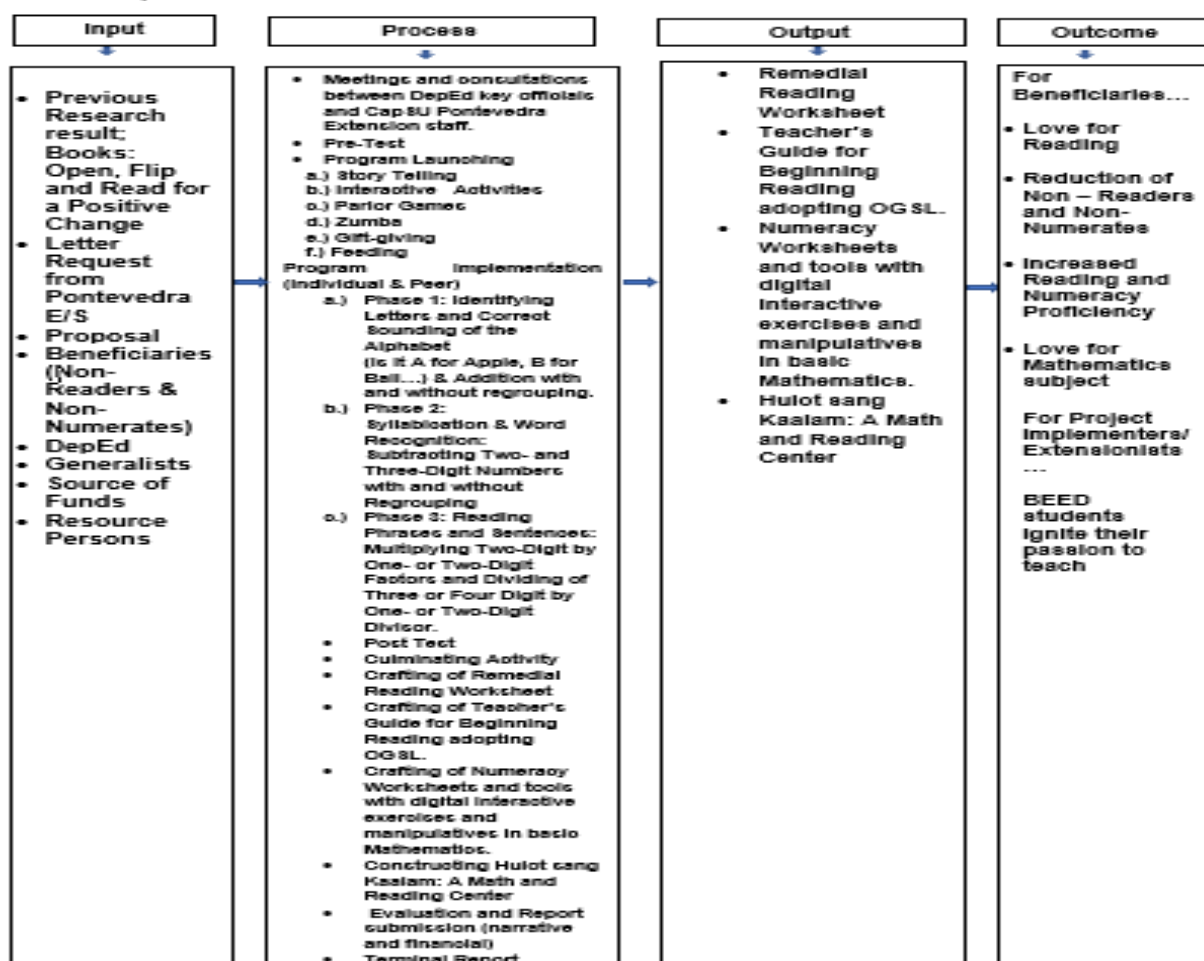
IV. Methodology

A letter from the office of the principal of Pontevedra Elementary School was received asking for any literacy program that can address their problem with regards to their non-readers and non-numerates. Planning was then conducted with the DepEd key officials headed by the schools division superintendent and the education supervisors of both the English and Mathematics department. There were 148 beneficiaries under this program.

A pre-test was conducted to the identified non-readers and non-numerates taken from the secondary data of the schools' Phil IRI result and BoSY CRLA

result. Post test was conducted to the beneficiaries and likewise, the secondary data of the Capiz Division Reading Assessors was utilized to assess the reading proficiency of the beneficiaries after the intervention done. Differentiated instruction such as individual and peer tutoring, games, and storytelling were employed. The OGSL of McGahan & Broadway (1975) was also employed. This is a multi-sensory, developmental, and therapeutic program designed to help learners construct a functional and secure language base that supports speaking, reading, writing and spelling. As a language arts base program, OGSL puts emphasis on teaching nonreaders and beginners in sounding the letters, blending sounds to form syllables, combining syllables to form words, and putting words together to build phrases, sentences, and short storylines.

Conceptual Framework



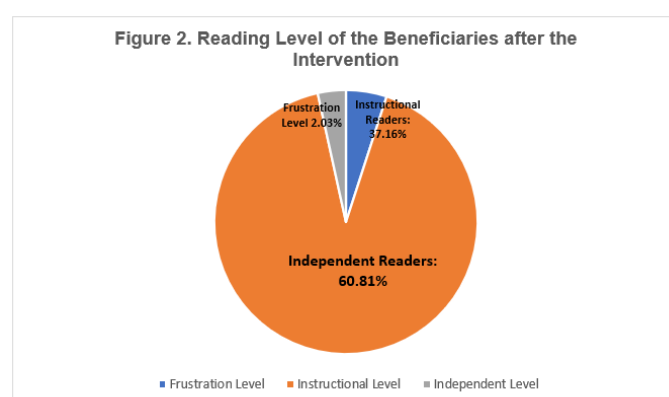
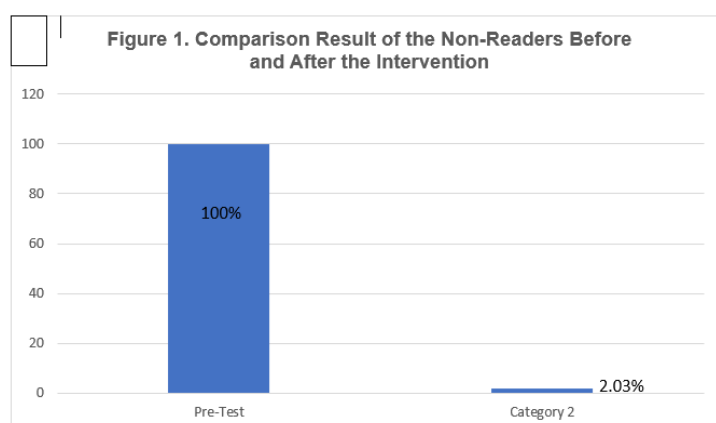
The everyday tutorial sessions to 148 beneficiaries were conducted by the BEED students during the duration of the extension project. This was done every morning on the remediation schedule incorporated in the class program of Grade 1 to Grade 6 of Pontevedra Elementary School through adopt -a pupil scheme employing differentiated instruction in enhancing their numeracy skills and OGSL strategy in teaching beginning reading to enhance their reading skills. The numeracy and reading worksheets for both the primary and intermediate students were both crafted as the outputs of the study. A teacher's guide in teaching beginning reading utilizing OGSL strategy was also crafted

while a reading and mathematics center called as “Hulot sang Kaalaman” was constructed by the recipient school.

IV. Results of Interventions

The tables and figures below illustrate the data comparison before and after the intervention program of the College of Education Arts & Sciences (COEAS) with regards to the Beginning Reading and Numeracy Proficiency of the beneficiaries. The study revealed that before the intervention there was 148 or 100% identified non-readers from Grade 1 to Grade 6 who were recipients of the tutorial sessions.

Furthermore, the data elucidated that after the conduct of the intervention there is only 3 or 2.03 % who was left in the frustration level which means that they refuse or cannot read at all, 55 or 37.16% in the instructional while 90 or 60.81% pupils in the independent level which means that they can read without the help of the teachers. This implies that excessive tutorial sessions through rigid contact time and with appropriate remedial tools and utilizing appropriate strategy like OGSL can improve the love for reading of pupils and enhance their reading skills.



Moreover, it can be noticed that there is a significant difference on the pre-test and post test result of the numeracy proficiency of the beneficiaries. Thus, delved from the data that Grade 1 pupils who were on the approaching stage ($M: 63.54$ or $SD: 28.11$) on the pre-test, they became advanced ($M: 92.36$ or $SD: 12.51$) on the post test. The Grade 2 pupils were on the developing stage ($M: 59.42$ or $SD: 34.11$) in the pre-test, they became proficient ($M: 59.42$ or $SD: 34.11$) during the post test. On the other hand, the Grade 3 pupils who were on the approaching proficiency ($M: 65.89$ or $SD: 26.70$) during the pre-test became advanced on the post test. While the Grade 4 and Grade 5 pupils, they were on the developing stage ($M: 56.20$ or $SD: 1.82$) & ($M: 54.75$ or $SD: 12.29$) during the conduct of the pre-test while on post test the Grade 4 pupils became proficient ($M: 84.40$ or $SD: 3.89$) and the Grade 5 became on the approaching proficiency stage ($M: 67.10$ or $SD: 4.09$).

Lastly, the Grade 6 pupils who were on the stage of approaching proficiency ($M: 63$ or $SD: 1.64$) during the conduct of the pre-test, after the intervention they became advanced ($M: 90.30$ or $SD: 2.34$). This implies that the everyday one-on-one tutorials using differentiated instructions, numeracy worksheets and other internet sources to the beneficiaries had great impact to them.

Table 1. Pre-Test Result of the Beneficiaries

Grade Level	No. of Non-Numerates	Mean	SD	Verbal Interpretation
1	22	63.54	28.11	Approaching Proficiency
2	23	59.42	34.11	Developing
3	41	65.89	26.70	Approaching Proficiency
4	22	56.20	1.82	Developing
5	23	54.75	2.29	Developing
6	15	63	1.64	Approaching Proficiency
Total	148			

Interpretation is based on the scale: Below 50% (Beginning); 50.01% - 62.50% (Developing); 62.51% - 75.00% (Approaching Proficiency); 75.01% - 87.50% (Proficient); 87.51% - 100% (Advanced)

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Table 2. Post Test Result of the Beneficiaries

Grade Level	Mean	SD	Verbal Interpretation
1	92.36	12.51	Advanced
2	84.42	21.66	Proficient
3	88.78	16.21	Advanced
4	84.40	3.89	Proficient
5	67.10	4.09	Approaching Proficiency
6	90.30	2.34	Advanced

Interpretation is based on the scale: Below 50% (Beginning); 50.01% - 62.50% (Developing); 62.51% - 75.00% (Approaching Proficiency); 75.01% - 87.50% (Proficient); 87.51% - 100% (Advanced)

Table 3. Test of Difference on the Numeracy Proficiency of the Beneficiaries

Test Statistics						
	prepostcater.1 - prepostgr.1	prepostcater.2 - prepostgr.2	prepostcatgr.3 prepostgr.3	prepostcater.4 - prepostgr.4	prepostcater.5 - prepostgr.5	prepostcatgr.6 - prepostgr.6
Asymp. Sig. (2-tailed)	-3.463* 0.001	-2.560* 0.010	-5.283* 0.000	-6.226 ^b .000	-5.588 ^b .000	-5.250 ^b .000

a. Wilcoxon Signed Ranks Test

b. Based on positive ranks.

Hence, the crafted teacher's guide in teaching beginning reading utilizing OGSL with the appropriate worksheets on reading is useful in the enhancement of the reading ability of the pupils. Likewise, the designed Mathematical tools both the worksheets and digital tools are of great aid on improving the numeracy proficiency of the pupils.

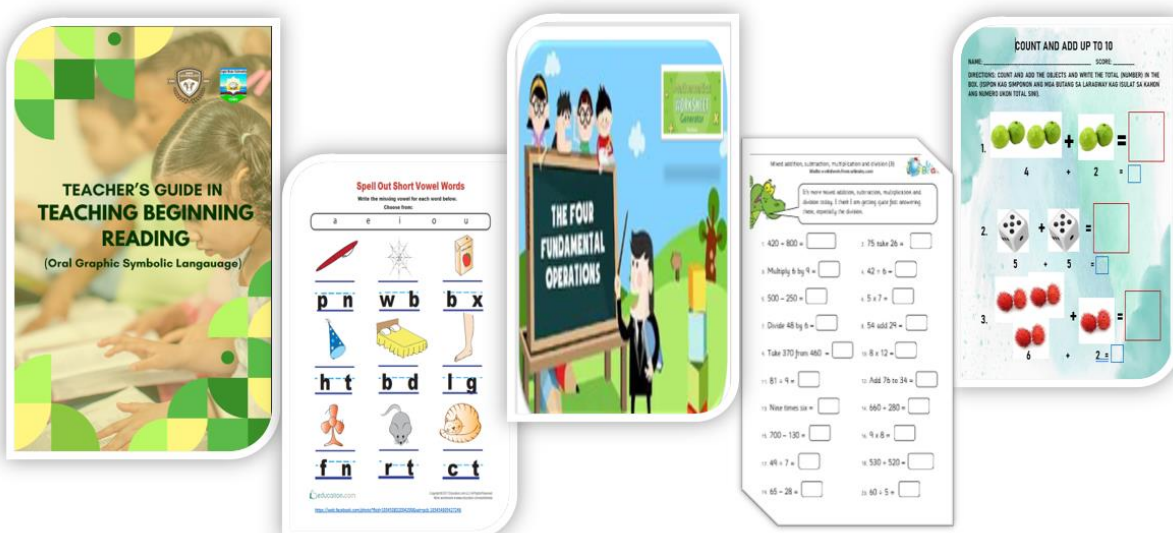
As to the impact of the intervention conducted to the adopted school, noticeably the reading level of the pupils in Pontevedra Elementary School had improved for most of them became independent readers, thus, they can read without the help of their teachers and parents. They too developed their loved for reading. Likewise, on their numeracy proficiency they improved from developing and approaching

Table 4. Matrix on the Result of the study.

Objectives	Methodology	Findings	Impact
1. Conduct tutorial sessions both in reading and numeracy to the adopted school; 2. Enhance the reading and numeracy level of the beneficiaries;	One-on-one tutorial sessions	- Reading Capability DepEd Secondary Data (Phil. IRI Result) Pre-Assessment: 148 or 18.83% non-readers (out of 786 graders-total population) Post Assessment 3 or 2.03% (Non-Readers) Extension Impact Assessment Pre-Test Result: 45.72 % Post Test Result : 93.16% Improved Reading Level: Frustration Level: 2.03% Instructional Level: 37.16% Independent Level: 60.81% Numeracy Proficiency DepEd Secondary Data (BoSY CRLA) Pre-Assessment: 148 or 18.83% non-readers (out of 786 graders-total population) Post Assessment: 68.96% :Proficient Extension Impact Assessment Pre-Test Result: 63.14% Approaching Proficiency Stage Post Test: 85.32% Proficient Stage	Loved for Reading Improved Reading Level Improved Numeracy Proficiency Enhance mathematical skills

Furthermore, as the outputs of the study, an **Teacher's Guide on Teaching Beginning Reading** utilizing Oral Graphic Symbolic Language (GSL) was crafted based on DepEd proposed strategies. Reading worksheets were also designed from the various reading tools, from Gawad Kalinga (GK), and other internet sources. This is especially designed to cater the needs of the non-readers and slow readers to become proficient independent beginning readers. This consists of reading lessons on the the basic of phonology starting from correct sound production to word recognition then reading of phrases and sentences and simple story lines. It has a guide on the proper way of pronouncing the vowels and consonant blends. This can be used by every remedial reading teacher as an instructional material and can even be utilized by the parents at home to help the non-readers and slow readers enhance both their beginning reading capability. This is a handy pamphlet that can be brought by pupils to their homes in order to have rigid practice to make them responsible young readers. On the other hand, numeracy worksheets on the four fundamentals of Mathematics and digital tools were specially designed to make every pupil to be motivated to engage in various activities in Mathematics and they may apply it in their everyday living.

Plate 1: The crafted Teacher's Guide in Teaching Beginning Reading and the crafted Worksheets in Reading and Mathematics



V. Lessons Learned (Conclusions)

1. Contact time with the non- readers and non-numerates is of very important to sustain the pupils' enthusiasm to read and be engaged in varied numeracy activities.
2. Exposing the pupils to varied reading activities motivate them to love reading.
3. Techniques in teaching reading and numeracy are essential.
4. Utilizing reading tools that fits on the level of reading ability of a pupil is necessary.

5. Utilizing appropriate worksheets and digital tools in Mathematics improved the numeracy proficiency of the pupils.

VI. Recommendations

1. Adopt the designed teacher's guide in teaching beginning reading.
2. Adopt the crafted worksheets both in reading and mathematics.
3. Mathematics teachers may adopt the designed digital tools in teaching Mathematics
4. Extend the tutorial sessions to help eradicate non-readers and make all pupils be in proficient and advanced stage.
5. Conduct a study on the reading ability of the pupils considering their reading difficulty.
6. Encourage the adopted school to utilize on their everyday reading and numeracy remediation the teacher's guide in beginning reading, the reading and mathematics worksheets and the digital tools in basic Mathematics.
7. Replicate the study to other public elementary school.

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